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Paleomagnetism of Permian sedimentary rocks from Tanzania and the Permian paleogeography of Pangea

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ABSTRACT

We report new paleomagnetic results from Karoo (Permian) rocks in Tanzania. K1 and K3 redbeds and K5 siltstones and limestones from the Songwe–Kiwira Basin, Ndeke redbeds from the Mikumi Basin, and group VIII redbeds from the Nyakatitu Basin were sampled and thermally demagnetized. Palynologic data and lithologic correlations suggest that the K1 redbeds are Lower Permian, and that the K3, Ndeke, and group VIII redbeds, as well as the K5 siltstones and limestones, are Upper Permian. Stable magnetic directions were obtained from the K3, Ndeke, and group VIII redbeds, yielding an Upper Permian pole at 26°N, 267°E. Characteristic remanence directions were also obtained from the K5 siltstones and limestones. However, because these directions define a pole (66°N, 269°E) that is very close to the Upper Cretaceous pole for Africa and significantly different from the Upper Permian redbed poles, these rocks may have been remagnetized. The K1 redbeds did not yield stable results.

Using our new results together with published Permian poles for Africa and the other Gondwanan continents, we revise the mean Lower and Upper Permian poles for Africa, West Gondwana, and Gondwana, and investigate whether or not these revised poles can discriminate between the various Pangea reconstructions proposed for the Permian with any greater certainty than previous (unrevised) mean poles. When compared to the mean Lower and Upper Permian poles for Laurussia, we find that our revised West Gondwanan and Gondwanan poles do not necessarily provide stronger support for any particular Pangea reconstruction than the previous poles. Therefore, even though this study improves the paleomagnetic data base for Gondwana for the Permian, it is not clear that the combination of our results with other paleomagnetic data from the Gondwanan continents yields significant new insights about the Permian configuration of Pangea. As suggested previously, strong non-dipole fields during the Permian may provide an explanation for why it is not easy to discriminate with certainty between the various Permian Pangea reconstructions, even with new paleomagnetic results.

Introduction

For more than two decades now a fundamental problem in Late Paleozoic to Early Mesozoic tectonics has been the configuration of the continents within Pangea from the Early Permian through the Late Triassic. While there is broad consensus that Pangea formed in the Late Carboniferous, there is ongoing debate regarding the exact configuration of the supercontinent between the Late Carboniferous and the Late Triassic. Paleomagnetic data from the Laurentian and Gondwanan continents for the Late Triassic and Early Jurassic are consistent with the classic Bullard et al. [1] reconstruction of Pangea (referred to as Pangea A1), and thus this reconstruction is widely accepted for the period just prior to the breakup of the supercontinent [e.g., 2–4]. However, the Pangea A1 reconstruction is not consistent with paleomagnetic data for the Permian and Early Triassic, motivating several authors to propose alternative reconstructions for the end-Paleozoic supercontinent, Pangea A2 [5], Pangea B [2], Pangea C [6], and Pangea D [4]. These various reconstructions represent two competing ideas, (1) an essentially static Pangea (Pangea A2) vs. (2) a mobile Pangea (Pangea B, C, and D). In the latter case, the models propose that Gondwana is sheared anticlockwise relative to Laurussia.

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Studies which have tried to test the various Pangea reconstructions using available paleomagnetic data show that it is difficult to discriminate with confidence between them because there are few well constrained Gondwanan paleopoles for the Permian and Early Triassic [3,4,7,8]. This study was undertaken to obtain new Permian poles for Africa that would improve the paleomagnetic data base for Gondwana and that could be used to test further the various Pangea reconstructions. Here we report new paleomagnetic results from Karoo (Permian) redbeds, siltstones, and limestones in Tanzania, revise Upper and Lower Permian paleopoles for Africa, West Gondwana, and Gondwana, and investigate whether or not these revised paleopoles provide new insights about the Permian configuration of Pangea.

The type section for the Karoo Formation in Tanzania is divided into eight stratigraphic units, K1–K8 [9]. In previous paleomagnetic work on the Tanzanian Karoo section, Opdyke [10] analyzed K3 redbed samples collected in a number of Karoo basins. Our study provides a more detailed paleomagnetic analysis of the Tanzanian Karoo section than Opdyke's in that we examine samples from K1, K3, K4, and K5 units, or their stratigraphic equivalents, which we also collected in several Karoo basins. Additionally, new palynologic data from Tanzanian Karoo rocks have become available enabling us to place tighter age constraints on Permian paleopoles from Tanzania than was possible previously.

Geology and field sampling

Karoo sediments in Tanzania are preserved in several basins downfaulted into metamorphosed basement of the Proterozoic Mozambique and Ubendian Belts (Fig. 1). Over 3000 m of predominately terrestrial sediments ranging in age from Late Carboniferous to Early Jurassic are found within these basins. Early work on the Tanzanian Karoo section can be found in Stockely and Oates [11], Stockely [9], Spence [12], McKinely [13], Harkin [14], Spence [15], Halderman [16], and McKinely [17]. More recently, Karoo sediments in Tanzania have been studied by Kreuser [18,25],

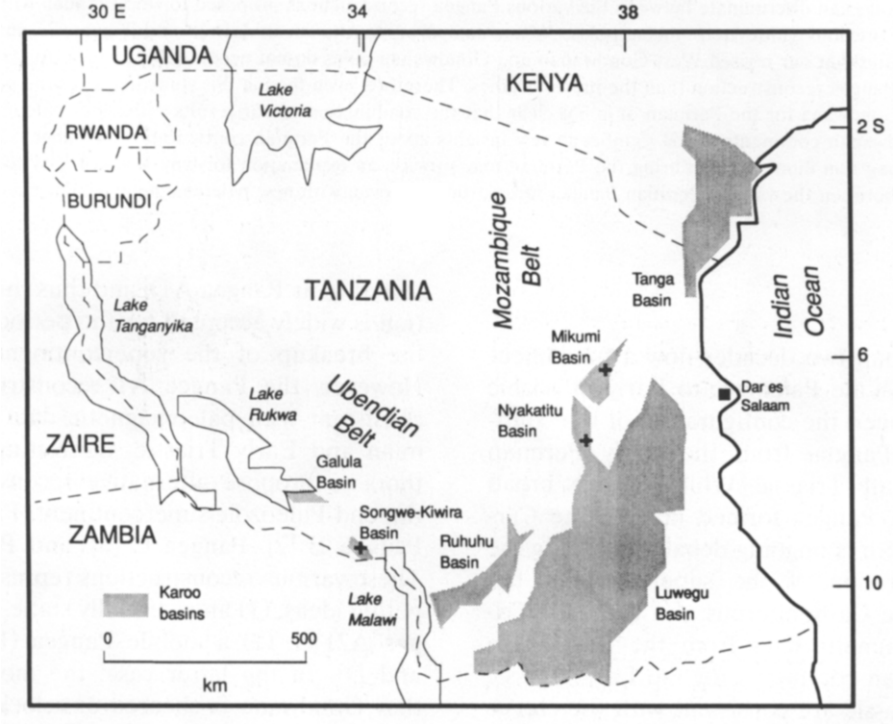


Fig. 1. Map showing the location of the major Karoo basins and sampling locations (+) in Tanzania.

Hankel [19], Kreuser and Semkiwa [20], Casshyap et al. [21], Kreuser et al. [22,23], and Dypvik et al. [24].

For paleomagnetic analysis 135 oriented cores were collected from 14 sites in the Mikumi, Nyakatitu, and Songwe–Kiwira Basins (Fig. 1) during July, 1990. Table 1 summarizes the stratigraphy in these basins, and below we briefly discuss their geology together with the sampling locations.

Mikumi Basin

The basal units in the Mikumi Basin, the "Ndeke Beds", are comprised of coarse siliciclastic sediments overlain by red mudstones and sandstones containing thin bands of calcitic and dolomitic nodules. Higher in the section, the Ndeke Beds are overlain by the "Calcareous Beds", a sequence of calcareous sandstones, limestones and carbonaceous shales capped by a thin marine sequence; the Calcareous Beds are in turn overlain by sandstones, carbonaceous shales, and mudstones (some red) referred to as the "Ruhembe Beds".

Samples were collected at three sites in the Ndeke redbeds (sites 1–3, Table 2) within 1–2 km of the Mikumi game park lodge. Red sandstones were sampled in stream beds at sites 1 and 2 (Table 2), and samples at site 3 were collected from red calcareous nodules exposed in a road bed drainage ditch. Sampling sites were separated by tens of meters stratigraphically, and the beds dipped to the west at angles of 25–35°. Because this basin has not been affected significantly by Cenozoic rift faulting, structures are likely related to downfaulting of the sediments into Precambrian basement during the Karoo period.

TABLE 1

Karoo stratigraphy

ERA	Stage	Songwe-Kiwira [14]	Mikumi [15]	Nyakatitu [15]
PERMIAN	Changxingian 248			
	Longtanian 250	K5	Ruhembe Beds	VIII
	Capitanian 253	K4 (missing)		VII
	Wordian 255		Calcareous Beds	VI
	Ufimian 256	K3		V
	Kungurian 260		Ndeke Beds	IV
	Artinskian 269	K2		III
	Sakmarian 282		Precambrian Basement	II
	Asselian 290	K1		I
Carboniferous				

Note Lower Permian = Asselian–Kungurian; Upper Permian = Ufimian–Changxingian. Time scale after Harland et al. [38]; numbers give ages in Ma.

Nyakatitu Basin

The Nyakatitu Basin is separated from the Mikumi basin by thick alluvial sediments which are probably Pliocene. Sediments in the Nyakatitu basin can be correlated lithologically with units in the Mikumi Basin, although the basin stratigraphy is labeled differently. Sandstones and mudstones comprise the lowermost groups (I and II, Table 1), which are equivalent to the Ndeke Beds in the Mikumi Basin. The intermediate groups (III–V) are calcareous beds capped by a thin marine unit, similar to the Mikumi Calcareous Beds. The uppermost groups (VI–VIII) correlate lithologically with the Ruhembe Beds, consisting predominately of sandstones, shales, and mudstones, some of which are red.

Samples were collected at two sites in the group VIII red mudstones and siltstones (sites 4 and 5, Table 2). Outcrops, dipping at angles of 20–35° to the west, were found in gullies along the western bank of the Ruaha River east of the village of Msolwa. Site 5 is separated from site 4 by at least 10 m stratigraphically. Similar to the Mikumi Basin, basin structures probably developed at the time of downfaulting of the sediments into Precambrian basement during the Karoo period.

Songwe–Kiwira Basin

The stratigraphic description of the Karoo sequence in the Songwe–Kiwira Basin follows the

classification of the Karoo sediments in the Ruhuhu Basin (the Tanzania Karoo type section), comprising eight units labeled K1 to K8 [9,11,14]. The lowermost unit (K1) consists of a diamictite member which is overlain by a varved sequence and followed by a sequence of shales, siltstones and red sandstones. The K2 unit, comprising sandstones, shales, siltstones, and coal, conformably overlies the K1 unit. Resting unconformably on the K2 unit are light-colored sandstones and red and purple siltstones and mudstones of the K3 unit. The K4 unit is missing, and K5 calcareous siltstones and limestones are found at the top of the sequence.

Paleomagnetic samples were collected at nine sites in the Songwe–Kiwira Basin. K1 red sandstones were sampled underground in the Kiwira Coal Mine (site 6, Table 2). K3 red mudstones were sampled at five stratigraphic horizons exposed in a road cut along the access road to the Kiwira Coal Mine (sites 7–11, Table 2); each horizon was separated by five or more meters stratigraphically. K5 calcareous siltstones were sampled in the Mwega River southeast of the

TABLE 2

Paleomagnetic data

Location (Formation) and site No.	n	D_o/I_o	D/I	κ	α_{95}	Site Location Latitude (S), Longitude (E)
Mikumi (Ndeke beds)						7.33°, 37.18°
1	7	115°/37°	128°/57°	104	6°	
2	13	125°/37°	140°/62°	109	4°	
3	5	121°/41°	125°/60°	95	8°	
Nyakatitu (group VIII beds)						7.77°, 37.10°
4	9	118°/17°	124°/56°	20	12°	
5	8	102°/25°	102°/59°	25	11°	
Songwe–Kiwira (K1)						9.48°, 33.67°
6	18	−/−				
Songwe–Kiwira (K3)						9.48°, 33.67°
7	12	144°/69°	97°/63°	115	4°	
8	8	173°/56°	123°/52°	87	7°	
9	7	174°/65°	110°/57°	60	8°	
10	6	171°/61°	133°/62°	43	10°	
11	6	124°/62°	100°/50°	462	3°	
Songwe–Kiwira (K5)						9.20°, 34.07°
12	9	7°/−41°	350°/−55°	64	7°	
13	6	−/−				
14	7	357°/−51°	344°/−38°	25	12°	
Songwe–Kiwira (K3; Opdyke, [10])						
	8		95°/58°	29	11°	
	4		126°/46°	312	5°	
Ketewaka–Mchuchuma (K3; Opdyke, [10])						
	7		113°/57°	176	5°	
	8		130°/55°	777	2°	
Galula (K3; Opdyke, [10])						
	8		157°/73°	37	9°	
	5		166°/65°	404	4°	
	7		177°/73°	83	7°	
	5		185°/66°	326	5°	
	9		172°/68°	125	5°	

n = No. of samples used for determining mean remanence directions; D_o/I_o = mean in-situ declination/inclination; D/I = mean declination/inclination after tilt correction; κ = precision parameter; α_{95} = 95% cone of confidence.

Kiwira Coal Mine (site 12, Table 2), and K5 limestones were sampled along stream beds in the Kigali and Lolela Hills near the village of Ngana, also southeast of the Kiwira Coal Mine (sites 13 and 14, Table 2). Where sampled, beds dipped to the east at 20–30°. Some of the basin structures may be related to the Cenozoic development of the Lake Malawi and Lake Rukwa rifts [26], and therefore it is not clear whether the attitudes of the units sampled resulted from tectonic activity associated with basin preservation during the Karoo or from faulting related to Cenozoic rifting.

Age of the Karoo Formation

The ages of Karoo units in the Mikumi, Nyakatitu and Songwe–Kiwira Basins have been determined biostratigraphically or by lithologic correlation with Karoo sequences in other areas of Tanzania and/or Africa. Starting with the oldest unit (K1), the diamictites and varved sediments are interpreted as Permo-Carboniferous glacial deposits related to the pan-Gondwana glaciation that is well documented elsewhere in Africa as Late Carboniferous to Early Permian [27–30]. Palynomorph assemblages from K1 shales in the Mchuchuma sub-basin (Ruhuhu Basin; Fig. 1) also suggest a Lower Permian age [Asselian; 31].

Originally, palynomorph assemblages from the Mchuchuma sub-basin suggested an Artinskian–Kungurian age for the K2 unit [32–34], but Wopfner and Kreuser [31] have recently placed K2 beds in the Ruhuhu Basin in the Sakmarian or Lower Artinskian. Nevertheless, the palynologic data clearly suggest that the K2 units are Lower Permian. K2 rocks can be correlated lithologically between the Ruhuhu and Songwe–Kiwira Basins, and with the Lower Permian Ecca Group in South Africa and other Lower Permian coal-bearing units in Zimbabwe, Zambia, and Zaire [29,35–37].

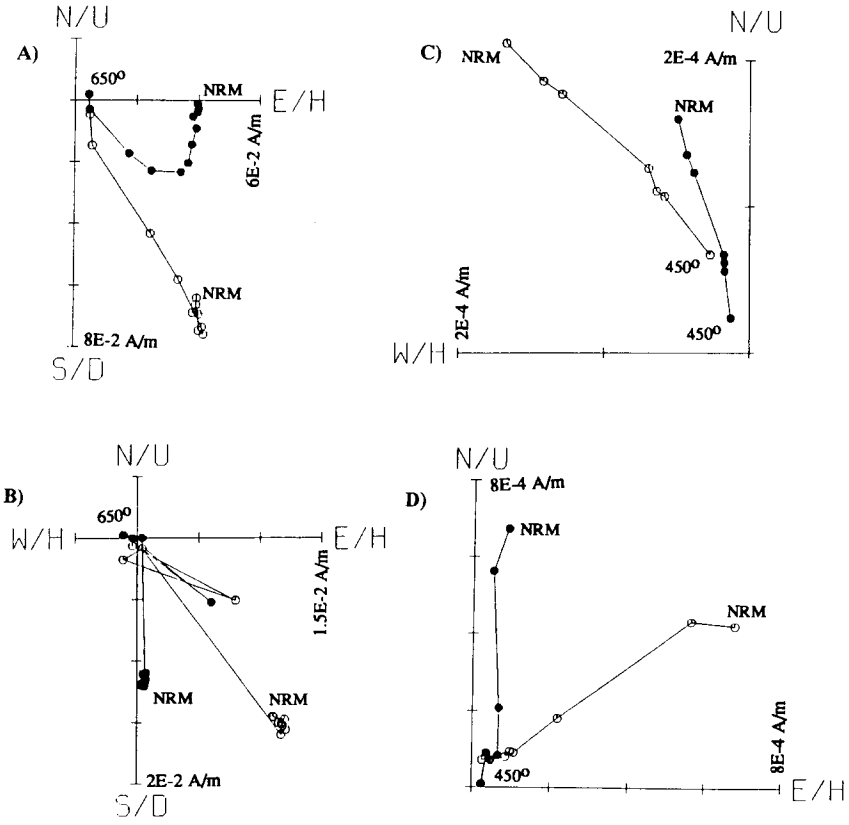


Fig. 2. Orthogonal plots of the magnetization vector (in situ) with increasing demagnetization temperature for representative samples (*N*, *E*, *S*, *U*, *H*, *D* = north, east, south, up, horizontal, and down, respectively). (A) Ndeke redbed sample. (B) K1 redbed sample. (C) K5 calcareous siltstone sample. (D) K5 limestone sample.

The K3 unit has not yielded any diagnostic pollen record so far (Kreuser, pers. commun.). However, because the K3 unit rests unconformably on K2 beds and is overlain by Upper Permian K4 and/or K5 sediments (see below), the K3 unit is probably early Upper Permian. The Ndeke Beds in the Mikumi Basin can be correlated lithologically with K3 beds in the Songwe–Kiwira and Ruhuhu Basins, and therefore they are probably early Upper Permian as well.

Palynomorph assemblages from the Ruhuhu basin place the K4 and K5 units in the Upper Permian (Kreuser, pers. commun.), consistent with biostratigraphic data from the Luwegu Basin (Fig. 1) which place K5 beds in the Upper Permian [19]. The K4 and K5 units in the Ruhuhu and Luwegu Basins correlate lithologically with the Ruhembe Beds in the Mikumi Basin, where palynologic data also yield an Upper Permian age [31]. Lithologic correlation of the Ruhembe Beds and the K4 and K5 units in the Ruhuhu and Luwegu Basins with the Nyakatitu group VIII beds and the Songwe–Kiwira K5 sediments suggests that these rocks are also Upper Permian.

In summary, we sampled K1, K3, K4, and K5 units or their stratigraphic equivalents in three Karoo basins. With respect to the units sampled, a review of the biostratigraphic data and lithologic correlations suggests that in the Songwe–Kiwira Basin the K1 redbeds are Lower Permian, and that the K3 redbeds and the K5 siltstones and limestones are Upper Permian. The Ndeke redbeds in the Mikumi Basin are stratigraphically equivalent to the K3 unit and are consequently Upper Permian, and the group VIII redbeds in the Nyakatitu Basin are stratigraphically equivalent to K4/K5 units and are thus Upper Permian as well. Following the time scale of Harland et al. [38], our best numerical age estimates are ~ 270–290 Ma for the Songwe–Kiwira K1 redbeds, ~ 254–260 Ma for the Songwe–Kiwira K3 and Mikumi Ndeke redbeds, and ~ 245–254 Ma for the Nyakatitu group VIII redbeds and the Songwe–Kiwira K5 siltstones and limestones.

Paleomagnetism

Thermal demagnetization was applied to all samples; specimens were stepwise heated and then cooled in field free space ($< 5\ \gamma$). Magnetic

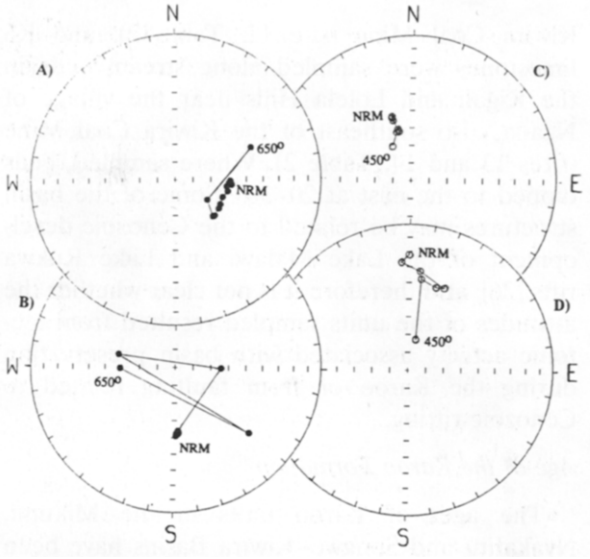


Fig. 3. Lower hemisphere stereonet plot of remanence directions at different demagnetization temperatures (after tilt correction) for the samples shown in Fig. 2. (A) Ndeke redbed sample. (B) K1 redbed sample. (C) K5 calcareous siltstone sample. (D) K5 limestone sample. ● = north-seeking direction down; ○ = south-seeking direction down.

remanence measurements were made using a Superconducting Technology cryogenic magnetometer. Orthogonal and stereonet plots are shown in Figs. 2 and 3 for representative samples. Except for the K1 samples, redbed specimens generally yielded stable magnetic directions. Figures 2a and 3a give the results from a typical well-behaved redbed sample, yielding a stable, southeasterly and steeply down (reversed) remanence direction that is characteristic of the K3, Ndeke, and group VIII redbeds. Results for a magnetically unstable K1 redbed sample are shown in Figs. 2b and 3b. This sample has a consistent, southerly direction up to 500°C, but at higher temperatures intensities decline rapidly and remanence directions become erratic. Maximum unblocking temperatures for the redbed samples were above 600°C, suggesting that hematite is the principal magnetic carrier.

The K5 calcareous siltstones and limestones were demagnetized to 450–500°C. Representative samples from sites 12 and 14 are shown in Figs. 2c,d and 3c,d. Samples were weak magnetically and intensities decreased steadily with increasing temperature. Directions are northerly (normal polarity) and do not change significantly with

demagnetization. By contrast, demagnetization of samples from site 13 did not yield stable magnetic directions.

Table 2 summarizes the pertinent information for each site. Remanence directions for each sample were determined using principal component analysis [39]. The remanence directions in Table 2 are mean values after removing those samples from a site whose directions were further than two angular standard deviations from the site mean direction. After tilt correction, the Songwe–Kiwira K3, the Mikumi Ndeke, and the Nyakatitu group VIII sites yield a similar southeasterly and steeply down direction.

The K3 and Ndeke redbeds probably lie within the Kiaman reversed polarity interval, which ended in the early Upper Permian [40]. Elsewhere, mixed polarities have been reported later in the Upper Permian [41–44], and because a total of eight stratigraphic horizons in the K3 and Ndeke redbeds yielded reversed polarities, it is unlikely that these units could be younger than Kiaman. The reversed directions obtained from the Nyakatitu sites also suggests that these redbeds may lie within the Kiaman Supercron, but because only two horizons were sampled in the Nyakatitu Basin, it is also possible that they could lie within a reversed polarity interval later in the Upper Permian.

For comparison, results from Opdyke's study [10] are shown at the bottom of Table 2. Our results from the Songwe–Kiwira K3, the Mikumi Ndeke, and the Nyakatitu group VIII redbeds are in close agreement with Opdyke's Songwe–Kiwira and Ketewaka–Mchuchuma K3 results. Opdyke's K3 results from the Galula Basin, however, are considerably different; remanence directions for the Galula site are more southerly and somewhat steeper than for the sites mentioned above. We discuss this observation in the next section.

Two observations suggest that the remanence directions isolated from the K3, Ndeke, and group VIII redbeds are primary; (1) most of the samples yielded a consistent, stable direction at high temperatures that was different from magnetic directions obtained at lower temperatures, and (2) the pole positions calculated from these stable magnetic directions do not resemble any Mesozoic or younger African paleopoles. Additionally, different structures in the Mikumi and Songwe–Kiwira

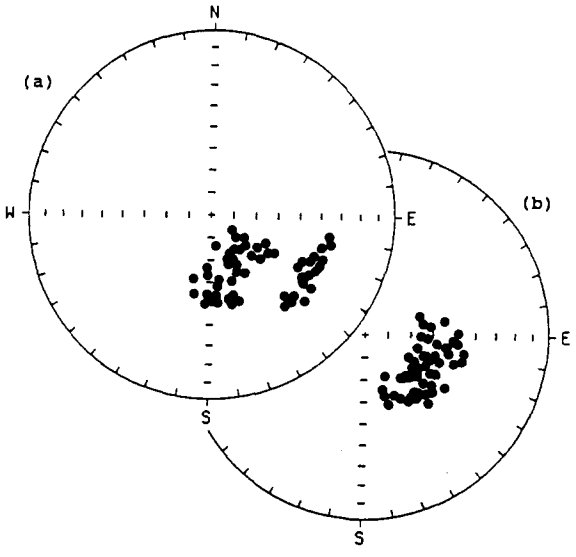


Fig. 4. Tilt test for Songwe–Kiwira K3 redbed and Mikumi Ndeke redbed samples. (a) Before tilt correction. (b) After tilt correction.

Basins permit a tilt test, which the data pass at a 99% confidence level. The tilt test is illustrated in Fig. 4. In situ, sample directions fall into two groups. Samples from Mikumi have shallower, easterly directions, while the Songwe–Kiwira samples have steeper and somewhat more southerly directions. With tilt correction, sample directions move together increasing the precision parameter (κ) from 17 to 43, and decreasing the 95% cone of confidence (α_{95}) from 5 to 3°. As mentioned above, the age of the basin structures is not known with certainty, particularly for the Songwe–Kiwira Basin. However, because the Mikumi Basin has not been clearly disrupted by the Cenozoic rift faulting common in other areas of Tanzania, structures in this basin probably developed in the Triassic or Jurassic during later stages of basin formation. If this is the case, then the positive tilt test suggests that the remanence directions obtained from samples in both the Mikumi and Songwe–Kiwira Basin pre-date any tectonic activity that affected the Mikumi Basin in the Triassic or Jurassic. If remanence directions in the Songwe–Kiwira Basin post-dated the tectonic activity in the Mikumi Basin, it seems unlikely that upon tilt correction directions from this basin would cluster more tightly with directions from the Mikumi Basin.

Remanence directions obtained from the K5 siltstones and limestones are different from the K3, Ndeke and group VIII redbed directions. We are suspicious that the magnetic directions isolated from the K5 siltstones and limestones are not primary directions. Although many of the siltstones and limestones gave stable results, the directions define a pole position (66°N, 249°E after tilt correction) that lies near the Upper Cretaceous pole for Africa (67°N, 245°E; [8]), suggesting that these rocks have been remagnetized. A structure test to check for post-tilting remagnetization of the K5 samples was inconclusive.

Discussion

In our discussion we first revise the Permian paleopoles for Africa, West Gondwana, and Gondwana, and then investigate whether or not these revised poles discriminate between the various Pangea reconstructions with any more certainty than previous (unrevised) West Gondwanan and Gondwanan paleopoles.

Table 3 summarizes the ages and positions for the Tanzanian poles, together with other Permian poles for Africa, which come from Morocco and Algeria. We do not include results from the K5 siltstones and limestones because these rocks may have been remagnetized. The results from the Mikumi, Songwe–Kiwira, and Ketewaka–Mchuchuma Basins are similar, providing a well constrained African pole for the early Upper Permian. This early Upper Permian (K3/Ndeke) pole is also similar to the Upper Permian Nyakatitu pole, however, they are both significantly different from the Lower Permian Moroccan/Algerian poles and the Galula K3 pole. The difference between the K3/Ndeke and Nyakatitu poles and the Moroccan/Algerian poles can be reasonably attributed to a difference in age (Upper vs. Lower Permian), but it is more difficult to explain the discrepancy between the K3/Ndeke and Nyakatitu poles and the Galula K3 pole by an age difference, as others have done [3,5,8,10], because palynologic data obtained recently from oil well cores in the Rukwa Rift near Galula suggest that much of the Galula Karoo section is Upper Permian [50].

If the Galula K3 redbeds are similar in age to

TABLE 3

Permian poles for Africa [a]

Formation/Location	Age	n'	Pole [b] (Lat, Lon)	κ'	α'_{95}	Pole [c] (Lat, Lon)	Reference
Tanzania							
redbeds, Nyakatitu	Pu	2	22°, 268°	–	–	32°, 266°	This study
Ndeke redbeds, Mikumi	Pu	3	34°, 262°	167	10°	43°, 259°	This study
K3 redbeds, Songwe–Kiwira	Pu	5	22°, 266°	37	13°	32°, 264°	This study
K3 redbeds, Songwe–Kiwira	Pu	2	22°, 271°	–	–	32°, 270°	Opdyke (1964)
K3 redbeds, Ketewaka	Pu	2	28°, 266°	–	–	38°, 264°	Opdyke (1964)
mean for K3/Ndeke redbeds	Pu	12	26°, 266°	41	7°	36°, 264°	
K3 redbeds, Galula	Pu	5	46°, 220°	227	5°	49°, 209°	Opdyke (1964)
Morocco							
Redbeds, Djebel Tarhat	Pl	11		35	8°	24°, 244°	Martin et al. (1978)
Redbeds, Chougrane	Pl	1		20	5°	32°, 244°	Daly and Pozzi (1976)
Trachyandesites, Taztot	Pl	1		80	5°	39°, 237°	Daly and Pozzi (1976)
Algeria							
Redbeds, Abadla	Pl	13		59	6°	29°, 240°	Morel et al. (1981)

[a] Only paleopoles having a quality factor (Q) of 3 or greater are shown (see Van der Voo [48] for a discussion of the paleomagnetic Q scale). Q values from Van der Voo [8] were used for published African poles.

[b] Pole positions in local coordinates.

[c] Pole positions in northwest African coordinates using rotation parameters from Lottes and Rowley [49] to correct for post-Permian rotations within Africa.

n' = number of site poles averaged; κ' = precision parameter for site poles; α'_{95} = 95% cone of confidence for site poles; Lat, Lon = Latitude (N), Longitude (E).

TABLE 4

Permian paleopoles for the Gondwana continents [a]

Age	Africa				South America				India				Australia			
	Lat, Lon	n'	κ'	α'_{95}	Lat, Lon	n'	κ'	α'_{95}	Lat, Lon.	n'	κ'	α'_{95}	Lat, Lon	n'	κ'	α'_{95}
Pu	34°, 265°	5	183	5°	61°, 236°	5	77	9°	43°, 224°	1	–	–	34°, 261°	3	37	21°
Pl	31°, 241°	4	114	9°	36°, 234°	3	399	6°					28°, 242°	1	–	–

[a] Pole positions are given in northwest African coordinates using the rotation parameters of Lottes and Rowley [49]. Lat, Lon = Latitude (N) and Longitude (E); κ', α'_{95}, and n' are defined in Table 3.

the K3 redbeds elsewhere in Tanzania, then why is there a significant discrepancy in the pole positions? A possible explanation is that the Galula site has been affected by structural rotation during the Cenozoic. The Karoo sediments at Galula lie on a Precambrian basement block in the Rukwa Rift that is flanked on either side by major basin bounding faults. Structural studies show that the Rukwa Rift formed from oblique extension [26,51], and therefore rotation of basement blocks within or adjacent to the rift, including the Galula block, may have occurred. Because the Galula site lies in a region of complex structure, we have chosen not to include it in our revision of Permian poles for Africa.

Table 4 gives mean Upper and Lower Permian paleopoles for the Gondwanan continents. The Upper Permian pole for Africa is the mean of the Nyakatitu group VIII, Mikumi Ndeke, Songwe–Kiwira K3, and Ketewaka–Mchuchuma K3 poles. The Lower Permian pole for Africa in Table 4 is the mean of the Moroccan/Algerian poles in Table 3, and the mean poles for South America, India, and Australia are from Van der Voo [8]. The Lower Permian poles for Africa, South America, and Australia are in fairly good agreement; in contrast, the Upper Permian poles do not cluster tightly. The reason for the differences between the Upper Permian poles is not clear; there are several possible explanations, such as age differences, remagnetizations, inappropriate reconstruction parameters, or non-dipole field effects.

Mean Permian poles for West Gondwana (Africa and South America) and Gondwana have been calculated from the paleopoles in Table 4 and are presented in Table 5 (revised poles). The Lower Permian poles are fairly well constrained ($\alpha_{95} = 5°$), whereas the larger uncertainty ($\alpha_{95} =$ 10 and 11°) in the Upper Permian poles reflects the greater differences in the Upper Permian poles for the Gondwanan continents. For comparison, unrevised Permian poles, which do not include our new African poles, are also given in Table 5.

Four Pangea reconstructions (A2, B, C, and D) have been proposed for the Permian and Early Triassic. In the Pangea A2 reconstruction (Fig. 5a), northwestern South America fits tightly into the present day Gulf of Mexico and juxtaposes northwest Africa against the eastern US seaboard. Pangea B places Africa beneath Europe and South America beneath eastern North America (Fig. 5b). Pangea C is a more extreme version of Pangea B, with South America placed beneath Europe and Africa beneath southern Asia, and Pangea D is similar to Pangea B but with Florida situated adjacent to northwestern Africa and Iberia placed north of northern Africa.

Given the generally held view that the Lauren-

TABLE 5

Mean Permian paleopoles for Gondwana [a]

Age	Revised West Gondwana Pole				Revised Gondwana Pole				Unrevised West Gondwana Pole				Unrevised Gondwana Pole			
	Lat, Lon	n'	κ'	α'_{95}	Lat, Lon	n'	κ'	α'_{95}	Lat, Lon	n'	κ'	α'_{95}	Lat, Lon	n'	κ'	α'_{95}
Pu	47°, 256°	10	18	10°	46°, 255°	14	20	11°	61°, 236°	5	77	9°	48°, 247°	9	17	13°
Pl	32°, 239°	7	149	5°	33°, 238°	8	141	5°	34°, 242°	8	60	7°	33°, 242°	9	67	6°

[a] Paleopoles are in northwest African coordinates using the rotation parameters from Lottes and Rowley [49].
Lat, Lon = Latitude (N), Longitude (E); κ', α and n' are defined in Table 3.

tian and Gondwanan continents were assembled in the Bullard et al. [1] configuration (Pangea A1), prior to the break-up of the supercontinent in the Early Jurassic, then the alternative reconstructions (Pangea A2, B, C, and D) proposed for the Permian and Early Triassic imply that a transition took place along a transform or megashear zone to bring the continents into the Pangea A1 configuration sometime prior to the Early Jurassic. A transition from Pangea A2 to Pangea A1 requires relatively small displacements along a megashear, while a transition from Pangea B, C, or D to Pangea A1 requires much larger displacements, as much as several thousand kilometers. The proposed existence of such a megashear, whether it be associated with a few hundred (Pangea A2) or a few thousand (Pangea B, C, and D) kilometers of continental displacement, can be used to test the various reconstructions on geological grounds. Summaries of the evidence for and against the existence of a megashear can be found in Morel and Irving [52], Hallam [53], and Smith and Livermore [4], and while possible shear zones can be suggested that are compatible with either relatively small (i.e., Pangea A2) or large displacements (i.e., Pangea B, C, or D), the geological evidence generally favors a Pangea A2 type reconstruction. However, geological evidence, or the lack thereof, cannot be used to reject the Pangea B, C, and D reconstructions outright [4].

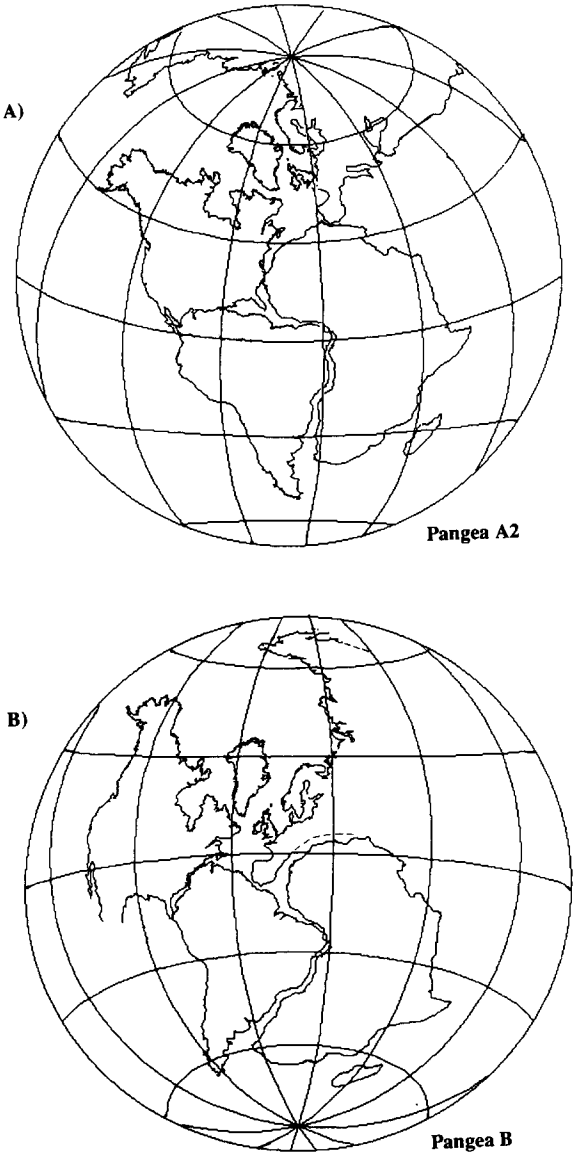


Fig. 5. Pangea reconstructions showing the North American, South American, African and Eurasian continents. (A) Pangea A2. (B) Pangea B.

Van der Voo [8] and Smith and Livermore [4] have recently attempted to test the Pangea reconstructions using available paleomagnetic data. Their results are somewhat inconclusive. They show that there are disagreements between the Permian Laurussian and Gondwanan poles for the Pangea A2 reconstruction, the reconstruction we favor on geological grounds, while the paleomagnetic data are in better agreement with the Pangea B, C, and D reconstructions, reconstructions that appear to us to be less plausible geologically. Figure 6a,b illustrates the disagreements in the paleomagnetic data for the Pangea A2 reconstruction. In the Lower Permian, there is no overlap in the 95% cones of confidence between either the West Gondwanan or Gondwanan poles and the Laurussian poles. Similarly, in the Upper Permian there is no overlap in the 95% cones of confidence for the West Gondwanan and Laurussian poles. For the Upper Permian Gondwanan and Laurussian poles there is some overlap in the cones of confidence, but that is only because there is a large uncertainty (α_{95}) in the Upper Permian Gondwanan pole.

Do our revised paleopoles in Table 5 provide more convincing support for the Pangea A2 reconstruction by improving the fit between the Laurussian and Gondwanan poles? To address that question, in Fig. 6c,d we have plotted our

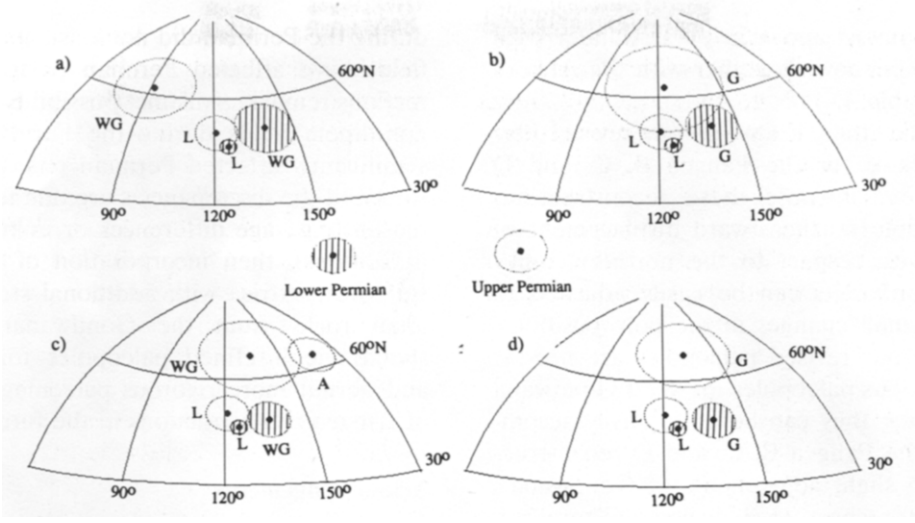


Fig. 6. Stereonet plots of Lower (Pl) and Upper Permian (Pu) paleopoles in North American coordinates using reconstruction parameters of Lottes and Rowley [49] for Pangea. (a) Previous mean poles for West Gondwana (*WG*) and Laurussia (*L*) (unrevised poles, Table 5). (b) Previous mean poles for Gondwana (*G*) and Laurussia. (c) Revised mean poles for West Gondwana, Laurussia, and Africa (*A*). (d) Revised mean poles for Gondwana and Laurussia. Circles show the 95% cones of confidence about the mean poles. Laurussian poles are from Van der Voo [56].

revised poles from Table 5 together with the Laurussian poles. It is clear that although the revised West Gondwanan poles may lie closer to the Laurussian poles than before, both the West Gondwanan and Gondwanan poles are still not in good agreement statistically with the Laurussian poles; the 95% cones of confidence of the poles do not overlap in either the Lower or Upper Permian. Thus, even with our revision of the Permian poles for West Gondwana and Gondwana, the paleomagnetic data do not strongly support the Pangea A2 reconstruction. This conclusion would also hold if other Pangea A2 configurations were used in Fig. 6 instead of the Lottes and Rowley [49] reconstruction [i.e., 54,55].

The Upper Permian pole for South America (Table 4) is similar to the Upper Triassic South American pole (67°N, 236°E in W. African coordinates; [8]), and therefore one might suspect that the Upper Permian South American rocks were remagnetized in the Upper Triassic. If true, then the mean Upper Permian West Gondwanan pole would comprise only the poles from Africa. The Upper Permian African pole (Table 4) has been plotted in Fig. 6c to determine if it is in any better agreement with the Laurussian pole than is the West Gondwanan pole. The African pole lies further from the Laurussian pole, and thus even by removing what may be suspect South American poles from the data set, the observation that the West Gondwanan and Laurussian poles are not in good agreement remains. The discrepancies between the new African poles and the Laurussian poles for the Pangea A2 reconstruction are further illustrated in Fig. 7, where the Car-

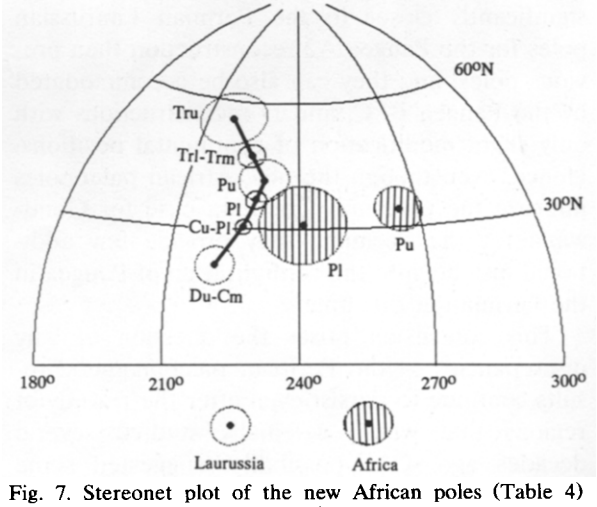


Fig. 7. Stereonet plot of the new African poles (Table 4) together with the Carboniferous–Triassic apparent polar wander path for Laurussia in northwest African coordinates using the reconstruction parameters of Lottes and Rowley [49] for Pangea. Circles show the 95% cones of confidence about the mean poles. Laurussian poles are from Van der Voo [56].

boniferous–Triassic apparent polar wander path for Laurussia is shown together with the African poles from Table 4.

What implications, if any, do our revised Permian poles have for the Pangea B, C, and D reconstructions? Because these reconstructions involve a simple northeastward displacement of Gondwana with respect to the northern continents, the continents can be easily adjusted to account for small changes in the pole positions. And because our revised paleopoles are not far from the previous paleopoles for West Gondwana and Gondwana, they can be quite easily accommodated in the Pangea B, C, and D reconstructions through slight adjustments of Gondwana's position with respect to Laurussia. Therefore, similar to the Pangea A2 reconstruction, our revised West Gondwanan and Gondwanan paleopoles do not discriminate between these alternative Pangea reconstructions with any more certainty than previous paleopoles.

Summary and conclusions

We have presented new Permian paleopoles for Africa and used them, together with other paleomagnetic data, to revise the mean Lower and Upper Permian poles for Africa, West Gondwana, and Gondwana. The revised West Gondwanan and Gondwanan paleopoles do not lie significantly closer to the Permian Laurussian poles for the Pangea A2 reconstruction than previous poles, and they can also be accommodated by the Pangea B, C, and D reconstructions with only slight modification of continental positions. Hence, even though the new African paleopoles improve the paleomagnetic data base for Gondwana for the Permian, they provide few additional insights into the configuration of Pangea in the Permian at this time.

This conclusion raises the question of why discrepancies in the Permian paleomagnetic results continue to persist even after the restudy of regions that where originally studied several decades ago. One possibility, suggested some twenty years ago by Briden et al. [57], is that the discrepancies in the Permian paleomagnetic data arise from large non-dipole components in the Permian geomagnetic field. Because there is little evidence indicating that large non-dipole fields during the Permian did not exist, that non-dipole fields have affected Permian paleomagnetic directions remains a distinct possibility. However, if non-dipole fields during the Permian have not significantly affected Permian paleomagnetic results and the discrepancies are due to some other reason (e.g., age differences or unknown remagnetizations), then incorporation of our new results from Africa with additional studies of Permian rocks from the Gondwanan continents should yield refined paleopoles for Gondwana and permit more rigorous paleomagnetic testing of Pangea reconstructions in the future.

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